

EDA: Bivariate & Multivariate Analysis

CampusX • 100 Days of Machine Learning • Day 21

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Field	Detail
Day	Day 21 — 100 Days of ML (CampusX)
Topic	EDA: Bivariate and Multivariate Analysis
Dataset Used	Seaborn tips iris titanic
Libraries	pandas, seaborn, matplotlib, numpy
Pre-requisite	Day 20: Univariate Analysis
GitHub Notebook	campusx-official/100-days-of-machine-learning



Figure 1 — Bivariate Analysis studies the relationship between two variables

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1. What is Bivariate Analysis?

Bivariate Analysis is the process of examining two variables simultaneously to understand the relationship or association between them. It answers questions like: Does studying more lead to better marks? Does a bigger house always cost more? It is the step after Univariate Analysis (Day 20) and before building ML models.

Analysis Type	Variables Studied	Goal
Univariate	1 variable	Understand the distribution of one column
Bivariate	2 variables	Find the relationship between two columns
Multivariate	3 or more variables	Find patterns across many columns at once

■ *Note: In the CampusX video, the tips dataset is used throughout. Each row is one restaurant visit. Columns include total_bill, tip, sex, smoker, day, time, and size.*

2. Types of Bivariate Analysis

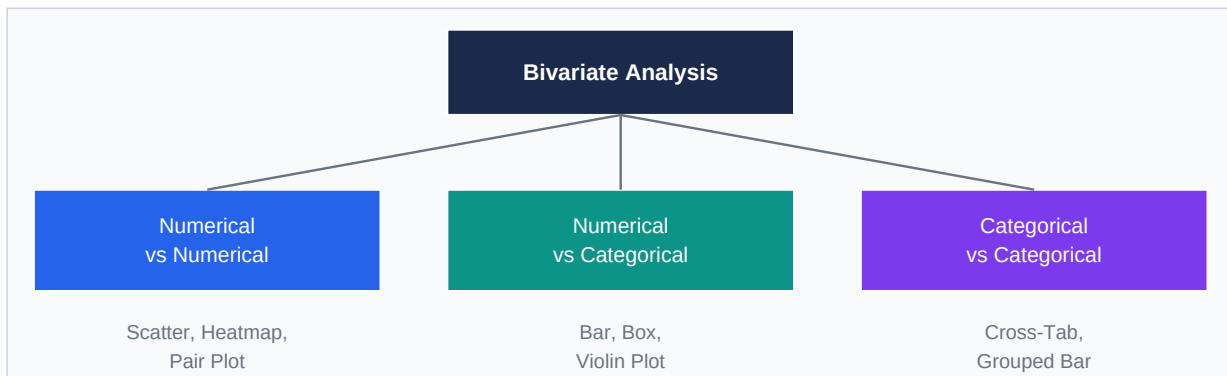


Figure 2 — Three main types based on the data type of each variable

Type	Variable X	Variable Y	Best Plots
Numerical vs Numerical	Number	Number	Scatter Plot, Correlation, Heatmap, Pair Plot
Numerical vs Categorical	Number	Category	Bar Plot, Box Plot, Violin Plot, Strip Plot
Categorical vs Categorical	Category	Category	Cross-Tab, Grouped Bar, Stacked Bar, Heatmap

The choice of plot always depends on the data type of the two variables. Getting this right is one of the most important EDA skills.

3. Numerical vs Numerical

When both variables are numbers, we want to see whether they move together, move in opposite directions, or have no relationship at all. The four main tools are: Scatter Plot, Correlation Coefficient, Heatmap, and Pair Plot.

3.1 Scatter Plot

A scatter plot places one variable on the X-axis and the other on the Y-axis. Each row in the dataset becomes a single dot. You can encode additional categorical variables using hue (colour), style (marker shape), and size.

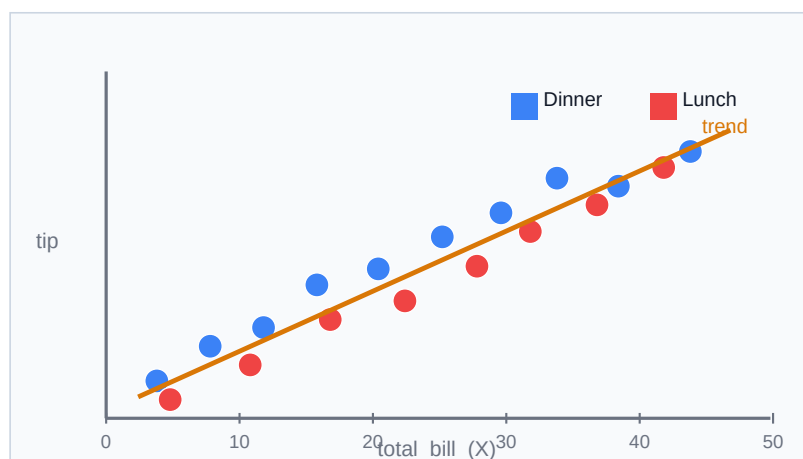


Figure 3 — Scatter plot of total_bill vs tip (blue = Dinner, red = Lunch, amber line = trend)

```
import seaborn as sns
import matplotlib.pyplot as plt
tips = sns.load_dataset('tips') # Basic scatter
sns.scatterplot(data=tips, x='total_bill', y='tip') plt.show() # With hue, style and size
sns.scatterplot(data=tips, x='total_bill', y='tip', hue='sex', # colour by sex
style='smoker', # shape by smoker
size='size' # dot size by party size )
plt.title('total_bill vs tip') plt.show()
```

✓ **Tip:** A clear upward trend means positive correlation. Random cloud-like spread means little or no correlation.

3.2 Pearson Correlation Coefficient

Correlation (r) measures the strength and direction of the linear relationship between two numerical variables. It always lies between -1 and +1.

$$r = \text{Cov}(X, Y) / (\text{SD}(X) \times \text{SD}(Y))$$

Value of r	Meaning	Example
+0.7 to +1.0	Strong positive	total_bill & tip both go up together
+0.3 to +0.7	Moderate positive	age & income — slight increase
0 to +0.3	Weak / no correlation	shoe size & exam score
-0.3 to -0.7	Moderate negative	speed & time taken

Value of r	Meaning	Example
-0.7 to -1.0	Strong negative	temperature & hot-drink sales

```
# Correlation between two columns tips['total_bill'].corr(tips['tip']) # Correlation
matrix (all numeric columns) tips.corr(numeric_only=True)
```

■ *Correlation only measures LINEAR relationships. Two variables can be strongly related in a curve but show r close to 0.*

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3.3 Correlation Heatmap

A heatmap visualises the entire correlation matrix in one glance. Dark blue cells indicate high positive correlation; white or light cells indicate weak correlation. The diagonal is always 1.0 (a variable is perfectly correlated with itself).

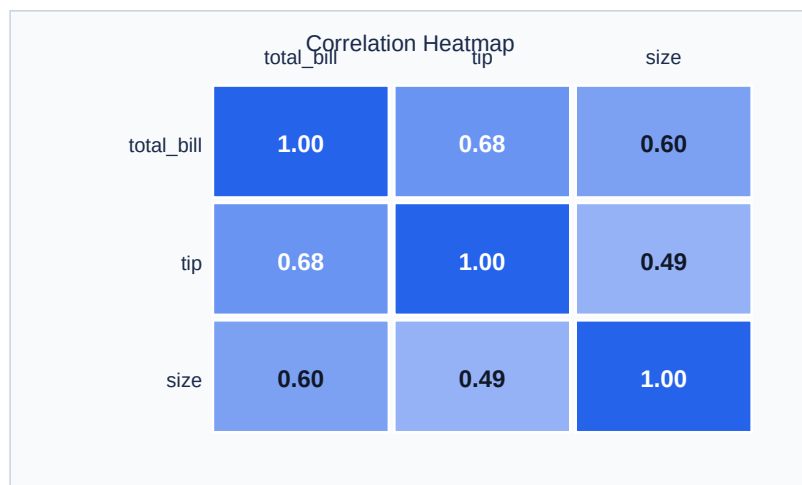


Figure 4 — Correlation heatmap for tips (total_bill & tip show $r = 0.68$ — moderate positive)

```
import seaborn as sns
import matplotlib.pyplot as plt

# Step 1: get the correlation matrix
corr = tips[['total_bill', 'tip', 'size']].corr()

# Step 2: draw the heatmap
plt.figure(figsize=(10, 8))
sns.heatmap(corr, annot=True, cmap='coolwarm', cbar=True)
plt.title('Correlation Heatmap for Tips')
plt.show()
```

3.4 Pair Plot

A Pair Plot contains a grid of plots. The diagonal shows histograms of each variable. The off-diagonal plots show scatter plots of pairs of variables, illustrating bivariate relationships in the data.



Figure 5 — Pair Plot grid (diagonal = histogram of each variable, off-diagonal = scatter between pairs)

```
# Basic pair plot
sns.pairplot(tips)
plt.show()

# With hue to colour by a category
sns.pairplot(tips, hue='sex')
plt.show()

# Only specific columns
sns.pairplot(tips[['total_bill', 'tip', 'size']])
plt.show()
```

```
tips[['total_bill', 'tip', 'size', 'sex']], hue='sex', diag_kind='kde' # KDE curve  
on the diagonal ) plt.show()
```

■ *Note: Pair plots become very slow and hard to read when you have more than 8-10 columns. Use .corr() heatmap instead for wide datasets.*

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4. Numerical vs Categorical

When one variable is a number and the other is a category, we want to see how the number changes across different groups. For example: does total_bill differ by day? Does tip change by sex?

4.1 Bar Plot (mean per group)

A bar plot shows the mean (average) of a numerical column for each category. Seaborn's barplot also draws a confidence interval bar automatically, which shows how reliable the mean estimate is.

```
# Mean total_bill for each day sns.barplot(data=tips, x='day', y='total_bill')
plt.title('Average Total Bill per Day') plt.show() # Grouped by two categories
sns.barplot(data=tips, x='day', y='total_bill', hue='sex') plt.show()
```

■ *Note: barplot shows the MEAN. If you want counts, use countplot instead.*

4.2 Box Plot (distribution per group)

A box plot shows the full distribution of a number across groups. It shows median, IQR (interquartile range), whiskers and outliers. It is more informative than a bar plot because it reveals spread, skewness and outliers.

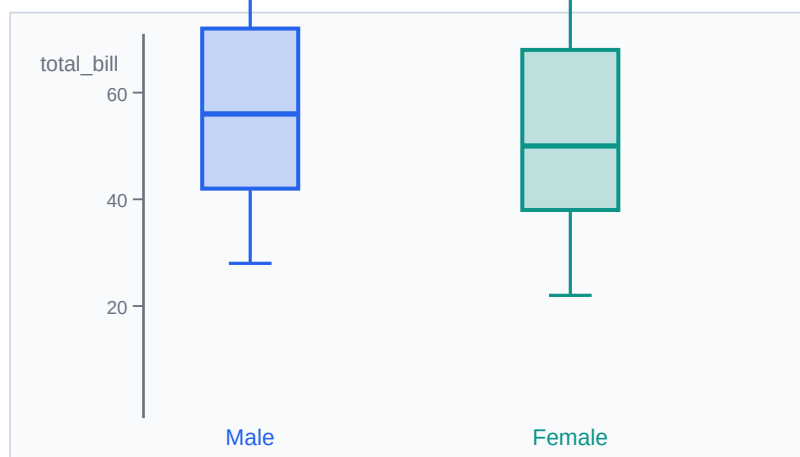


Figure 6 — Box plots of total_bill by sex (red dots = outliers, horizontal line = median)

```
# Box plot – total_bill by day sns.boxplot(data=tips, x='day', y='total_bill')
plt.show() # With hue sns.boxplot(data=tips, x='day', y='total_bill', hue='sex')
plt.show()
```

Box Plot Part	What it Shows
Middle line	Median (50th percentile)
Box bottom	Q1 (25th percentile)
Box top	Q3 (75th percentile)
Box height	IQR = Q3 - Q1 (middle 50% of data)
Whiskers	Q1 - 1.5 x IQR to Q3 + 1.5 x IQR
Dots outside	Outliers (beyond 1.5 x IQR)

4.3 Violin Plot

A violin plot combines a box plot with a KDE curve. The wider the violin at any point, the more data points are near that value. It shows the full shape of the distribution, not just summary statistics.

```
sns.violinplot(data=tips, x='day', y='total_bill') plt.show() # Split by sex on the
same violin sns.violinplot(data=tips, x='day', y='total_bill', hue='sex', split=True)
plt.show()
```

4.4 Strip Plot and Swarm Plot

Both plots show every individual data point — nothing is summarised. Strip plots add random horizontal jitter to avoid overlapping dots. Swarm plots spread points so they never overlap.

```
# Strip plot sns.stripplot(data=tips, x='day', y='total_bill', jitter=True, hue='sex')
plt.show() # Swarm plot (best for small datasets < 500 rows) sns.swarmplot(data=tips,
x='day', y='total_bill', hue='sex') plt.show()
```

✓ **Tip: Overlay a strip plot on top of a violin or box plot to show both the distribution shape and every individual point at once.**

5. Categorical vs Categorical

When both variables are categories, we want to count how often different combinations appear. For example: how many male vs female smokers visited on each day?

5.1 Cross-Tabulation (pd.crosstab)

A cross-tab is a frequency table. Rows are the values of one category, columns are the values of another, and cells are the counts.

```
# Count rows for each sex & day combination pd.crosstab(tips['sex'], tips['day']) #  
Normalise to proportions (0-1) pd.crosstab(tips['sex'], tips['day'], normalize='index')  
# rows sum to 1 # With a third variable as values pd.crosstab(tips['sex'], tips['day'],  
values=tips['tip'], aggfunc='mean')
```

Example output of `pd.crosstab(tips['sex'], tips['day'])`:

sex \ day	Fri	Sat	Sun	Thur
Female	9	28	18	32
Male	10	58	58	30

5.2 Grouped Bar Chart

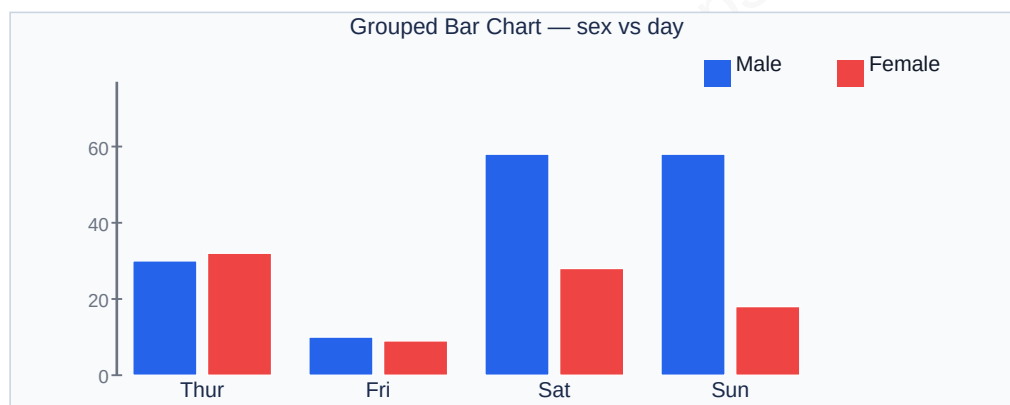


Figure 7 — Grouped bar chart: count of visits by day and sex

```
# sns.countplot counts rows automatically sns.countplot(data=tips, x='day', hue='sex')  
plt.title('Number of Visits per Day by Sex') plt.show() # Stacked bar using crosstab ct =  
pd.crosstab(tips['day'], tips['sex']) ct.plot(kind='bar', stacked=True, figsize=(7,  
4)) plt.title('Stacked: Visits per Day by Sex') plt.show()
```

5.3 Categorical Heatmap (crosstab + heatmap)

Plot the cross-tab as a colour heatmap for a visual frequency table. Darker cells mean more occurrences.

```
import seaborn as sns import pandas as pd import matplotlib.pyplot as plt ct =  
pd.crosstab(tips['sex'], tips['day']) sns.heatmap(ct, annot=True, fmt='d',  
cmap='YlOrRd') plt.title('Count Heatmap: sex vs day') plt.show()
```

6. Multivariate Analysis

Multivariate Analysis looks at three or more variables at the same time. We typically add extra variables through colour (hue), shape (style), size, and by creating grid plots (FacetGrid / pairplot).

6.1 Scatter with hue + style + size (3-4 variables)

```
# 4 variables in one plot: # X = total_bill, Y = tip, colour = sex, shape = smoker
sns.scatterplot( data=tips, x='total_bill', y='tip', hue='sex', style='smoker',
size='size', sizes=(20, 200) ) plt.title('4 Variables in One Scatter Plot') plt.show()
```

6.2 FacetGrid (small multiples)

FacetGrid creates a grid of subplots, one for each value of a category. It is the best way to compare distributions across many groups without cluttering a single chart.

```
g = sns.FacetGrid(tips, col='time', row='sex', height=3, aspect=1.2)
g.map_dataframe(sns.scatterplot, x='total_bill', y='tip') g.add_legend() plt.show() #
Another example: histogram per category g = sns.FacetGrid(tips, col='day', col_wrap=2,
height=3) g.map(sns.histplot, 'total_bill', bins=15) plt.show()
```

6.3 Pair Plot with hue (full multivariate summary)

```
sns.pairplot( tips, hue='sex', vars=['total_bill', 'tip', 'size'], diag_kind='kde',
plot_kws={'alpha': 0.6} ) plt.suptitle('Pair Plot – Tips Dataset', y=1.02) plt.show()
```

7. Choosing the Right Plot

Question to Answer	X Type	Y Type	Best Plot	seaborn Function
Do X and Y move together?	Num	Num	Scatter	scatterplot()
How correlated are all my numbers?	Num	Num	Heatmap	heatmap(corr)
All pairwise relationships?	Num	Num	Pair Plot	pairplot()
Average of number per group?	Cat	Num	Bar Plot	barplot()
Distribution of number per group?	Cat	Num	Box Plot	boxplot()
Full shape of distribution per group?	Cat	Num	Violin Plot	violinplot()
Every individual point per group?	Cat	Num	Strip / Swarm	stripplot() swarmplot()
How many in each combo?	Cat	Cat	Cross-Tab	pd.crosstab()
Bar count per category pair?	Cat	Cat	Grouped Bar	countplot(hue=)
Frequency table as colours?	Cat	Cat	Cat Heatmap	heatmap(crosstab)

8. Complete Code Reference

All code below uses the seaborn tips dataset. Run the setup cell once, then any snippet below independently.

Setup

```
import pandas as pd import numpy as np import seaborn as sns import matplotlib.pyplot
as plt tips = sns.load_dataset('tips') iris = sns.load_dataset('iris') titanic =
sns.load_dataset('titanic') sns.set_theme(style='whitegrid') print(tips.head())
```

All Numerical vs Numerical Plots

```
# --- 1. Scatter plot --- sns.scatterplot(data=tips, x='total_bill', y='tip')
plt.show() # --- 2. Correlation matrix --- print(tips.corr(numeric_only=True).round(2))
# --- 3. Heatmap of correlation matrix --- sns.heatmap(tips.corr(numeric_only=True),
annot=True, fmt='.2f', cmap='Blues') plt.show() # --- 4. Pair plot ---
sns.pairplot(tips, hue='sex') plt.show()
```

All Numerical vs Categorical Plots

```
# --- 1. Bar plot (mean) --- sns.barplot(data=tips, x='day', y='total_bill', hue='sex')
plt.show() # --- 2. Box plot --- sns.boxplot(data=tips, x='day', y='total_bill',
hue='sex') plt.show() # --- 3. Violin plot --- sns.violinplot(data=tips, x='day',
y='total_bill', hue='sex', split=True) plt.show() # --- 4. Strip plot ---
sns.stripplot(data=tips, x='day', y='total_bill', jitter=True, hue='sex', dodge=True)
plt.show()
```

All Categorical vs Categorical Plots

```
# --- 1. Cross-tabulation --- print(pd.crosstab(tips['sex'], tips['day'])) # --- 2.
Normalised cross-tab --- print(pd.crosstab(tips['sex'], tips['day'],
normalize='index').round(2)) # --- 3. Grouped bar (count) --- sns.countplot(data=tips,
x='day', hue='sex') plt.show() # --- 4. Categorical heatmap --- ct =
pd.crosstab(tips['sex'], tips['day']) sns.heatmap(ct, annot=True, fmt='d',
cmap='YlOrRd') plt.show()
```

9. Common Mistakes and Best Practices

Mistake	Why it is wrong	What to do instead
Using a scatter plot for Cat vs Num	Scatter expects two numbers; categories on X look wrong	Use boxplot or barplot
Using barplot when you want to see distribution	barplot only shows the mean — hides spread and outliers	Use boxplot or violinplot
Ignoring outliers on box plots	Outliers can be errors or genuine important data points	Investigate each outlier separately
Not using hue when comparing groups	You miss important group differences	Always add hue='category' when comparing groups
Using pairplot on 20+ columns	Plot becomes tiny and unreadable	Use heatmap(corr) for wide datasets
Confusing correlation with causation	High r does not mean X causes Y	Use domain knowledge; run experiments to prove causation
Not scaling axes equally on scatter	Can exaggerate or hide the true slope	Use plt.axis('equal') or check axis ranges

10. Quick Reference Cheat Sheet

Task	One-liner Code
Scatter: X vs Y	<code>sns.scatterplot(data=df, x='col1', y='col2')</code>
Scatter with groups	<code>sns.scatterplot(..., hue='cat', style='cat2')</code>
Correlation: two columns	<code>df['A'].corr(df['B'])</code>
Full correlation matrix	<code>df.corr(numeric_only=True)</code>
Heatmap of correlations	<code>sns.heatmap(df.corr(), annot=True, cmap='Blues')</code>
Pair plot	<code>sns.pairplot(df, hue='category')</code>
Bar plot (mean per group)	<code>sns.barplot(data=df, x='cat', y='num')</code>
Box plot (dist per group)	<code>sns.boxplot(data=df, x='cat', y='num')</code>
Violin plot	<code>sns.violinplot(data=df, x='cat', y='num')</code>
Strip plot	<code>sns.stripplot(data=df, x='cat', y='num', jitter=True)</code>
Count per category	<code>sns.countplot(data=df, x='cat')</code>
Grouped bar (count)	<code>sns.countplot(data=df, x='cat', hue='cat2')</code>
Cross-tabulation	<code>pd.crosstab(df['cat1'], df['cat2'])</code>
Normalised cross-tab	<code>pd.crosstab(df['A'], df['B'], normalize='index')</code>
Categorical heatmap	<code>sns.heatmap(pd.crosstab(df['A'], df['B']), annot=True)</code>
FacetGrid by category	<code>g = sns.FacetGrid(df, col='cat'); g.map(sns.histplot, 'num')</code>
4-variable scatter	<code>sns.scatterplot(..., hue='c1', style='c2', size='c3')</code>
Stacked bar	<code>pd.crosstab(df['A'], df['B']).plot(kind='bar', stacked=True)</code>
Set global theme	<code>sns.set_theme(style='whitegrid')</code>
Save figure	<code>plt.savefig('plot.png', dpi=150, bbox_inches='tight')</code>

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